



PHY1004W

Tuesday Afternoon Laboratories and Tutorials 2022

As a science graduate, it will be assumed by your future employer that you have certain skills. These will include problem solving skills, the ability to engage with apparatus that you may not have seen before, the ability to plan and execute an experiment or some type of investigation, the ability to collect, analyse and interpret data and very importantly the ability to communicate and present your findings either orally or in the form of a written report. The practical component of this course therefore forms an important part of PHY1004W.

PHY1004W laboratories and tutorials will take place on **Tuesday afternoons** (only) from **14:00 to 17:00** in the **Course I laboratory** in the RW James building.

The class will be split into two groups to ensure that the Course Lab I does not exceed COVID regulation capacity.

Group 1 must be in the lab from 14:00 to 15:25.

Group 2 must be in the lab from 15:35 to 17:00.

To ensure that each group uses the short time effectively learning material relating to lab activities will be available on the PHY1004W course Vula site in advance. Students must read the material before the in-person laboratory sessions to prepare for the activity.

The coordinators for the PHY1004W laboratory are:

1st semester: Dr **Trisha Salagaram**, Room 5.13, RW James Building, trisha.salagaram@uct.ac.za

2nd semester: TBA

Postgraduate demonstrators will also assist you in the laboratory and during tutorials.

Laptop: Bring your laptop to all Tuesday afternoon activities.

Laboratory manuals: There is no formal PHY1004W laboratory manual. Worksheets describing the lab activity will be available on Vula prior to the lab. You will, however, receive a copy of a workbook, “Introduction to Measurement in the Physics Laboratory: A Probabilistic Approach.”

Marks: Your laboratory and tutorial marks will be posted on the Vula Gradebook tool. Please check these marks regularly and report errors or omissions to the coordinator by email.

Student submissions and marked work: Reports, worksheets and computer code must be submitted for marking on Vula. More details will be given in each practical. Marked reports will be returned on Vula.

Four types of activities will take place as part of your laboratory course: (a) practical experiments; (b) measurement tutorials; (c) computational tutorials; and (d) theory tutorials. These will be delivered on Vula under the Practicals lesson tab.

(a) **Practical experiments:** During these sessions you will be asked to complete a practical task, which will often take the form of a problem that requires an experimental solution. A video of the experimental setup will be provided. You will collect data in during your allotted session in the laboratory. You will need to analyse the results, plot graphs and write up a full report which will be graded by the demonstrators. Your report must be word processed. The submission date for the report will be a few days after the experiment. Completed reports must be uploaded to Vula before the submission deadline. Resources about lab report writing will be available on the course Vula site. There are further instructions concerning lab reports in the manual “Introduction to Measurement in the Physics Laboratory.” Reports will be marked by the demonstrators, and returned via Vula.

(b) **Measurement tutorials:** You will receive a copy of the book “Introduction to Measurement in the Physics Laboratory: A Probabilistic Approach” which contains a series of interactive exercises dealing with measurement in physics. This book needs to be used for all lab afternoons when either a “measurement tutorial” or “practical experiment” is scheduled. During “measurement tutorials” you will be working through particular sections of the workbook, which often will be accompanied by some form of experimental work. You will be required to submit your work on Vula.

(c) **Computational tutorials:** There will be 5 computational tutorials during the year during which you will learn how to solve physics problems using the language VPython. These will be worksheet-based activities which will be graded by the demonstrators.

(d) **Theory tutorials:** The theory discussed in lectures will be supported by 4 problem solving tutorials. Details of the activity will be made available by the lecturer lecturing you at the time of a tutorial. During this time, you will solve a number of problems set by the lecturers. More details will follow later.

Laboratory tests: There will be a laboratory practical test at the end of each semester. More information will be given closer to the time.

PHY1004W laboratory programme for 2022

Date	Activity	Deliverable
15 Feb	Practical experiment: The Rolling Ball	Report
22 Feb	Measurement Tutorial 1: Introduction to measurement	Vula validation
1 March	Theory Tutorial 1	Solutions
8 March	Measurement Tutorial 2: Dealing with dispersion	Worksheet
15 March	Science Faculty Test Day	
22 March	Measurement Tutorial 3: Speed of Sound	Vula validation
29 March	<i>Vacation</i>	
05 April	Practical experiment: Free fall	Report
12 April	<u>Computational Tutorial 1</u>	Worksheet and code
19 April	Theory Tutorial 2	Solutions
26 April	Practical experiment: Least Squares and Springs	Worksheet
3 May	<u>Computational Tutorial 2</u>	Worksheet and code
10 May	Laboratory Practical Test 1	Report
17 May	Swot	
	<i>Long vacation</i>	
26 July	Practical experiment: Capacitors	Report
2 August	<u>Computational Tutorial 3</u>	Worksheet and code
16 August	Measurement Tutorial 4: Working with uncertainties	Worksheet
23 August	Theory Tutorial 3	Solutions
30 August	Practical experiment: Circuits	Practical Validation
13 September	<u>Computational Tutorial 4</u>	Worksheet and code
6 September	<i>Vacation</i>	
20 September	Practical experiment: The Oscilloscope	Worksheet
27 September	Theory Tutorial 4	Solutions
4 October	Practical experiment: RC Circuit	Report/Worksheet
11 October	<u>Computational Tutorial 5</u>	Worksheet and code
18 October	Laboratory Practical Test 2	Report
25 October	Swot	